Supervyse OPF SMBIOS-MDR Developer's Guide

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2024-04-25	Glenn Hsu	Conrad Yao	Initial version

1 Introduction

A guide for developers to customize a new interface on D-Bus interfaces and assign the value from smbios-mdr. We create a property on D-Bus interfaces named “CompanyName” and a string “company name” is assigned as its value from smbios-mdr. We can view the results via the Smbios.MDR_V2 service.

Basic flow:

- First, add an interface named “Test” with “CompanyName” property in phosphor-dbus-interfaces.
- Second, assign the string “company name” in smbios-mdr
- Finally, we can view the result on the D-Bus interfaces of SMBIOS.

2 Add an interface to phosphor-dbus-interfaces

2.1 Get source code

Assuming you already have this SDK and have set up your build environment, you should now be in the directory "<SDK>/build/<MACHINE_PLATFORM>/". Use the following command:

```
devtool modify phosphor-dbus-interfaces
```

The source code for phosphor-dbus-interfaces is located at:

```
<SDK>/build/<MACHINE_PLATFORM>/workspace/source/phosphor-dbus-interfaces
```

In the following document, <SOURCE> indicates the below path:

```
<SDK>/build/<MACHINE_PLATFORM>/workspace/source
```

For example:

```
<SOURCE>/phosphor-dbus-interfaces/yaml
```

equals to

```
<SDK>/build/<MACHINE_PLATFORM>/workspace/source/phosphor-dbus-interfaces/yaml
```

2.2 Add a new .yaml file

Add a new .yaml file to create a new interface.

LOCATION: <SOURCE>/phosphor-dbus-interfaces/yaml/xyz/openbmc_project/Inventory/Decorator

Assuming we add a new .yaml named "Test.interface.yaml",

```
ex.
<SOURCE>/phosphor-dbus-interfaces\
/yaml/xyz/openbmc_project/Inventory/Decorator/Test.interface.yaml
```

Add the following code to "Test.interface.yaml". This code includes a property named "CompanyName".

```
 description: >
    A test sample
 properties:
    - name: CompanyName
      type: string
      description: >
          A test field
```

2.3 Add building-related files

Add building-related files (meson.build) to generate corresponding C++ code from the .yaml file.

LOCATION: <SOURCE>/phosphor-dbus-interfaces/gen/xyz/openbmc_project/Inventory/Decorator

One meson.build needs to be added and another meson.build needs to be modified.

1. Add one meson.build
   Add a directory named "Test" at the following location:
   <SOURCE>/phosphor-dbus-interfaces/gen/xyz/openbmc_project/Inventory/Decorator

   Then we enter the directory "Test" and add a new meson.build
   After the above steps, the meson.build file can be found in the following location:
   phosphor-dbus-interfaces/gen/xyz/openbmc_project/Inventory/Decorator/Test/meson.build

   After creating the meson.build file, add the following code.

```
generated_sources += custom_target(
    'xyz/openbmc_project/Inventory/Decorator/Test__cpp'.underscorify(),
    input:
[ '../../../../../../yaml/xyz/openbmc_project/Inventory/Decorator/Test.interf
ace.yaml',  ],
    output: [ 'server.cpp', 'server.hpp', 'client.hpp',  ],
    command: [
        sdbuspp_gen_meson_prog, '--command', 'cpp',
        '--output', meson.current_build_dir(),
        '--tool', sdbusplusplus_prog,
        '--directory', meson.current_source_dir() / '../../../../../../yaml',
        'xyz/openbmc_project/Inventory/Decorator/Test',
    ],
)
```

2. Modify one meson.build
   Modify the file (meson.build) located at
   phosphor-dbus-interfaces\/gen/xyz/openbmc_project/Inventory/Decorator/meson.build

   Add the following code at the bottom of meson.build.

```
subdir('Test')
generated_others += custom_target(
   'xyz/openbmc_project/Inventory/Decorator/Test__markdown'.underscorify(),
    input:
[ '../../../../../yaml/xyz/openbmc_project/Inventory/Decorator/Test.interfac
e.yaml',  ],
    output: [ 'Test.md' ],
    command: [
        sdbuspp_gen_meson_prog, '--command', 'markdown',
        '--output', meson.current_build_dir(),
        '--tool', sdbusplusplus_prog,
        '--directory', meson.current_source_dir() / '../../../../../yaml',
        'xyz/openbmc_project/Inventory/Decorator/Test',
    ],
)
```

2.4 Build phosphor-dbus-interfaces

Use the following command to build phosphor-dbus-interface.

```
bitbake phosphor-dbus-interfaces
```

2.5 Verify the result of building

Enter the directory "<SDK>/build/<MACHINE_PLATFORM>" and use the following command to verify if server.hpp has been generated.

```
find . -wholename \
'*phosphor-dbus-interfaces*xyz/openbmc_project/Inventory/Decorator/Test/server.hpp'
```

Finally, we can use the class generated by phosphor-dbus-interfaces in smbios-mdr. The image below shows the result.

```
./tmp/sysroots-components/armv7ahf-vfpv4d16/phosphor-dbus-interfaces/usr/include/xyz/openbmc_project/Inventory/D
ecorator/Test/server.hpp
./tmp/work/armv7ahf-vfpv4d16-openbmc-linux-gnueabi/phosphor-dbus-interfaces/1.0+git999-r1/image/usr/include/xyz/
openbmc_project/Inventory/Decorator/Test/server.hpp
./tmp/work/armv7ahf-vfpv4d16-openbmc-linux-gnueabi/phosphor-dbus-interfaces/1.0+git999-r1/packages-split/phospho
r-dbus-interfaces-src/usr/src/debug/phosphor-dbus-interfaces/1.0+git999-r1/phosphor-dbus-interfaces-1.0+git999/g
en/xyz/openbmc_project/Inventory/Decorator/Test/server.hpp
./tmp/work/armv7ahf-vfpv4d16-openbmc-linux-gnueabi/phosphor-dbus-interfaces/1.0+git999-r1/packages-split/phospho
r-dbus-interfaces-dev/usr/include/xyz/openbmc_project/Inventory/Decorator/Test/server.hpp
./tmp/work/armv7ahf-vfpv4d16-openbmc-linux-gnueabi/phosphor-dbus-interfaces/1.0+git999-r1/package/usr/src/debug/
phosphor-dbus-interfaces/1.0+git999-r1/phosphor-dbus-interfaces-1.0+git999/gen/xyz/openbmc_project/Inventory/Dec
orator/Test/server.hpp
./tmp/work/armv7ahf-vfpv4d16-openbmc-linux-gnueabi/phosphor-dbus-interfaces/1.0+git999-r1/package/usr/include/xy
z/openbmc_project/Inventory/Decorator/Test/server.hpp
./tmp/work/armv7ahf-vfpv4d16-openbmc-linux-gnueabi/phosphor-dbus-interfaces/1.0+git999-r1/sysroot-destdir/usr/in
clude/xyz/openbmc_project/Inventory/Decorator/Test/server.hpp
```

The file highlighted in red is

```
<SDK>/build/<MACHINE_PLATFORM>/tmp/sysroots-components/armv7ahf-vfpv4d16/phosphor-
dbus-interfaces/usr/include/xyz/openbmc_project/Inventory/Decorator/Test/server.hpp
```

This file includes the line "virtual std::string companyName(std::string value);" in class "Test".

```
/** Set value of CompanyName */
virtual std::string companyName(std::string value);
```

The code in smbios-mdr can then include this header and assign a value using the above virtual function.

3 Assign a value from smbios-mdr

3.1 Get source code

Assuming you already have this SDK and have set up your build environment. You should now be in the directory "<SDK>/build/<MACHINE_PLATFORM>/". Use the following command:

```
devtool modify smbios-mdr
```

The source code of smbios-mdr is located at:

```
<SOURCE>/smbios-mdr
```

3.2 Add smbios-mdr/include/test_sample.hpp

Add a test_sample.hpp
LOCATION: <SOURCE>/smbios-mdr/include/
Add the following code to "test_sample.hpp"

```
#pragma once
#include "smbios_mdrv2.hpp"
#include <xyz/openbmc_project/Inventory/Decorator/Test/server.hpp>

namespace phosphor
{
namespace smbios
{
using testIntf = sdbusplus::server::object_t<
    sdbusplus::xyz::openbmc_project::Inventory::Decorator::server::Test>;

class TestSample :testIntf
{
  public:
 // Delete default constructor without arguments to
    // force using arguments when creating an object
    TestSample() = delete;
 // Using default destructor to ensure resources will released
    // when deleting an object
    ~TestSample() = default;
 // Disable the copy constructor to prevent unintended copying
 // Example:
 //     TestSample test1(10);
 //     TestSample test2 = test1;
    TestSample(const TestSample&) = delete;
 // Disable the copy assignment operator
    // (prevent assigning a value from one object to another)
 // Example:
 //     TestSample test1(10);
 //     TestSample test2;
 //       test2 = test1;
    TestSample& operator=(const TestSample&) = delete;
 // Enable move constructor
```

```
    // (Transfer the resources of test1 to test2, and set test1 to nullptr.)
 // Example:
 //     TestSample test1(10);
 //     TestSample test2(std::move(test1));
    TestSample(TestSample&&) = default;
 // Enable move assignment operator
    // (Transfer the resources of test1 to test2, and set test1 to nullptr.)
 // Example:
 //     TestSample test1(10);
 //     TestSample test2;
 //     test2 = std::move(test1);
    TestSample& operator=(TestSample&&) = default;

 // Define the constructor
    TestSample(sdbusplus::bus::bus& bus, const std::string& objPath, int index,
             uint8_t* smbiosTableStorage) :
 // Initialize the parent class and variables when creating this object
        testIntf(bus, objPath.c_str()),
        storage(smbiosTableStorage), index(index)
    {
 // Update information when creating this object
        testSampleInfoUpdate();
    }
    void testSampleInfoUpdate(void);

  private:
    std::string path;  // object path
    uint8_t* storage;
    int index;
    void testSampleCompanyName();
};

} // namespace smbios

}
```

3.3 Add smbios-mdr/src/test_sample.cpp

Add a test_sample.cpp
LOCATION: <SOURCE>/smbios-mdr/src/
Add the following code to “test_sample.cpp” for assigning “comapny name” to the propety “CompanyName”.

```
#include "test_sample.hpp"
#include "mdrv2.hpp"
#include <fstream>
#include <iomanip>
#include <iostream>
#include <sstream>

namespace phosphor
```

```
{
namespace smbios
{
void TestSample::testSampleInfoUpdate(void)
{
    testSampleCompanyName();
}

void TestSample::testSampleCompanyName()
{
    companyName("company name");
}

}
}
```

3.4 Modify CMakeLists.txt, adding src/test_sample.cpp

Find the line includes "SRC_FILES" in the CMakeLists.txt, add "src/test_sample.cpp" at the end of the line. The image below shows the result after modifying.

```
set (SRC_FILES src/mdrv2.cpp src/mdrv2_main.cpp src/cpu.cpp src/dimm.cpp
     src/system.cpp src/pcieslot.cpp src/tpm.cpp src/firmware.cpp src/test_sample.cpp)
```

3.5 Modify smbios-mdr/include/mdrv2.hpp

1. Add the following line to the header includes in smbios-mdr/include/mdrv2.hpp.

   ```
   #include "test_sample.hpp"
   ```

   The image below shows the result after adding the line to the header includes. The line numbers in your mdrv2.hpp file may differ from those in the image below.

   ```
   17 #pragma once
   18 #include "baseboard.hpp"
   19 #include "cpu.hpp"
   20 #include "dimm.hpp"
   21 #include "firmware.hpp"
   22 #include "pcieslot.hpp"
   23 #include "smbios_mdrv2.hpp"
   24 #include "system.hpp"
   25 #include "tpm.hpp"
   26 #include "test_sample.hpp"
   27
   28 #include <sys/stat.h>
   29 #include <sys/types.h>
   30 #include <unistd.h>
   31
   ```

2. Add the following line to the private members of the MDR_V2 class in mdrv2.hpp.

```
std::vector<std::unique_ptr<TestSample>> testSampleCollection;
```

The image below shows the result after adding.

```
std::unique_ptr<Tpm> tpm;
std::vector<std::unique_ptr<Firmware>> firmwareCollection;
std::vector<std::unique_ptr<TestSample>> testSampleCollection;
std::vector<std::unique_ptr<Baseboard>> baseboards;
```

3.6 Modify smbios-mdr/src/mdrv2.cpp

Add the following code to "MDR_V2::systemInfoUpdate()" in mdrv2.cpp. The code is used to add "testSample" objects.

```
testSampleCollection.clear();
num = 2;

for (int index = 0; index < num; index++)
{
std::string objName = "TestSample_" + std::to_string(index);
const char* defaultTestSamplePath=
"/xyz/openbmc_project/inventory/system/chassis/testSample";
std::string path(defaultTestSamplePath);
path += "/" + objName;
   testSampleCollection.emplace_back(
       std::make_unique<phosphor::smbios::TestSample>(
        bus, path, index,
        smbiosDir.dir[smbiosDirIndex].dataStorage
       )
   );
}
```

The above code is suggested to add before "pcies.clear();" in mdrv2.cpp. The image below shows the position.

Note: It is not a screen shot of the file.

```
1
2   void MDR_V2::systemInfoUpdate()
3   {
4       ...
5       dimms.clear();
6       ...
7       }
8
9   #endif
10
11      // start
12      testSampleCollection.clear();
13      num = 2;
14
15      for (int index = 0; index < num; index++)
16      {
17          std::string objName = "TestSample_" + std::to_string(index);
18          const char* defaultTestSamplePath= "/xyz/openbmc_project/inventory/system/chassis/testSample";
19          std::string path(defaultTestSamplePath);
20          path += "/" + objName;
21          testSampleCollection.emplace_back(
22              std::make_unique<phosphor::smbios::TestSample>(
23                  bus, path, index,
24                  smbiosDir.dir[smbiosDirIndex].dataStorage
25              )
26          );
27      }
28      // end
29
30      pcies.clear();
31      ...
32  }
```

3.7 Build smbios-mdr

Use the following command to build smbios-mdr.

```
bitbake smbios-mdr
```

4 Build image

After successfully build phosphor-dbus-interfaces and smbios-mdr, enter directory "<SDK>/build/<MACHINE_PLATFORM>/" and build the image. Use the following command to build the image.

```
bitbake obmc-phosphor-image
```

5 Verification

1. Use the following command to verify that "testSample" objects are displayed.

```
busctl tree xyz.openbmc_project.Smbios.MDR_V2
```

```
root@BMC000000000000:/var/lib/smbios# busctl tree xyz.openbmc_project.Smbios.MDR_V2
`-/xyz
  `-/xyz/openbmc_project
    |-/xyz/openbmc_project/Smbios
    | `-/xyz/openbmc_project/Smbios/MDR_V2
    |-/xyz/openbmc_project/inventory
    | `-/xyz/openbmc_project/inventory/system
    |   `-/xyz/openbmc_project/inventory/system/chassis
    |     |-/xyz/openbmc_project/inventory/system/chassis/motherboard
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/CPU_0
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/CPU_1
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/ProcessorModule_0
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/ProcessorModule_0
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot0
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot1
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot10
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot2
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot3
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot4
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot5
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot6
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot7
    |     | |-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot8
    |     | `-/xyz/openbmc_project/inventory/system/chassis/motherboard/pcieslot9
    |     `-/xyz/openbmc_project/inventory/system/chassis/testSample
    |       |-/xyz/openbmc_project/inventory/system/chassis/testSample/testSample_0
    |       `-/xyz/openbmc_project/inventory/system/chassis/testSample/testSample_1
    `-/xyz/openbmc_project/software
      |-/xyz/openbmc_project/software/UEFI_0
      `-/xyz/openbmc_project/software/bios
```

2. Use following commands to verify that the “CompanyName” property displays with “company name”.

```
busctl introspect xyz.openbmc_project.Smbios.MDR_V2
/xyz/openbmc_project/inventory/system/chassis/testSample/testSample_0
```

```
root@BMC000000000000:/var/lib/smbios# busctl introspect xyz.openbmc_project.Smbios.MDR_V2 /xyz/openbmc_project/i
nventory/system/chassis/testSample/testSample_0
NAME                                        TYPE      SIGNATURE RESULT/VALUE   FLAGS
org.freedesktop.DBus.Introspectable         interface -         -             -
.Introspect                                 method    -         s             -
org.freedesktop.DBus.Peer                   interface -         -             -
.GetMachineId                               method    -         s             -
.Ping                                       method    -         -             -
org.freedesktop.DBus.Properties             interface -         -             -
.Get                                        method    ss        v             -
.GetAll                                     method    s         a{sv}         -
.Set                                        method    ssv       -             -
.PropertiesChanged                          signal    sa{sv}as  -             -
xyz.openbmc_project.Inventory.Decorator.Test interface -         -             -
.CompanyName                                property  s         "company name" emits-change writable
```

```
busctl introspect xyz.openbmc_project.Smbios.MDR_V2
/xyz/openbmc_project/inventory/system/chassis/testSample/testSample_1
```

```
root@BMC000000000000:~# busctl introspect xyz.openbmc_project.Smbios.MDR_V2 /xyz/openbmc_project/inventory/syste
m/chassis/testSample/testSample_1
NAME                                        TYPE      SIGNATURE RESULT/VALUE   FLAGS
org.freedesktop.DBus.Introspectable         interface -         -              -
.Introspect                                 method    -         s              -
org.freedesktop.DBus.Peer                   interface -         -              -
.GetMachineId                               method    -         s              -
.Ping                                       method    -         -              -
org.freedesktop.DBus.Properties             interface -         -              -
.Get                                        method    ss        v              -
.GetAll                                     method    s         a{sv}          -
.Set                                        method    ssv       -              -
.PropertiesChanged                          signal    sa{sv}as  -              -
xyz.openbmc_project.Inventory.Decorator.Test interface -         -              -
.CompanyName                                property  s         "company name" emits-change writable
root@BMC000000000000:~#
```